

# Review, Implementation, and Comparative Analysis of Encoder–Decoder Models with and without Attention Mechanism

*Paper: Efficient Machine Translation with a BiLSTM-Attention Approach*

Wu & Xing, arXiv:2410.22335, October 2024

Dataset: Tatoeba English-French (OPUS, v2023-04-12)

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# Research Paper Review

## Problem Statement

Modern machine translation has been dominated by Transformer-based architectures since Vaswani et al. (2017). While Transformers deliver high translation quality, they are computationally expensive and memory-intensive, making them unsuitable for resource-constrained environments such as mobile devices, embedded systems, or low-bandwidth deployment scenarios.

Wu and Xing (2024) propose an alternative: a Seq2Seq model using a Bidirectional LSTM (BiLSTM) encoder combined with an attention-enhanced decoder. Their central claim is that this architecture can match or exceed Transformer-level translation quality on WMT14 benchmarks while occupying significantly less storage space and requiring fewer computational resources.

## Architecture: Encoder–Decoder with Attention

The model follows a classic encoder-decoder structure with the following key components:

| Component                | Description                                                                                                                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BiLSTM Encoder           | Reads the input sequence in both forward and backward directions simultaneously. Produces a sequence of hidden states capturing left and right context at every position.                                                |
| Learnable Initial States | Instead of zero-initializing $h_0$ and $c_0$ , the paper introduces trainable parameters for the initial hidden and cell states, allowing the model to learn a task-optimal starting point.                              |
| Projection Layer         | A linear layer merges the forward and backward final hidden states of the BiLSTM into a single vector suitable for decoder initialization.                                                                               |
| Attention Mechanism      | Bahdanau (additive) attention is applied at each decoder step. The decoder queries all encoder hidden states to compute a context vector, rather than relying on a single fixed bottleneck vector.                       |
| LSTM Decoder             | At each step, the decoder receives the previous output token embedding concatenated with the attention context vector. The final prediction is produced from a linear projection over [LSTM output, context, embedding]. |

## Type of Attention Used

The paper employs Bahdanau Attention (Additive Attention), introduced by Bahdanau, Cho, and Bengio (2015). The attention score between decoder hidden state  $s$  and encoder hidden state  $h$  is:

$$\text{score}(s, h) = v \cdot \tanh(W_1 \cdot h + W_2 \cdot s)$$
$$\alpha = \text{softmax}(\text{score}) \quad | \quad \text{context} = \sum(\alpha_i \cdot h_i)$$

This allows the decoder to dynamically attend to any part of the source sentence at each generation step, resolving the fixed-context bottleneck of vanilla Seq2Seq models. Unlike Luong (dot-product) attention, Bahdanau attention uses a learned alignment function, making it more expressive.

## Dataset Used

The paper conducts experiments on the WMT14 (Workshop on Machine Translation 2014) dataset, the standard benchmark for neural machine translation research. WMT14 includes:

- English-German (EN-DE):** approximately 4.5 million sentence pairs

- **English-French (EN-FR):** approximately 35.8 million sentence pairs

Results are reported using BLEU and ROUGE scores. In this implementation, the Tatoeba English-French corpus (OPUS, v2023-04-12) was used as a substitute for faster training on Google Colab, with 500 filtered sentence pairs for proof-of-concept demonstration.

## Key Contributions

- **Innovative BiLSTM Encoder:** Captures bidirectional context with learnable initial hidden and cell states, outperforming unidirectional LSTM encoders.
- **Attention-Enhanced Decoder:** Applies Bahdanau additive attention at every decoding step, enabling dynamic source alignment.
- **Model Size Efficiency:** Achieves competitive or superior BLEU scores on WMT14 with a significantly smaller parameter footprint than Transformers.
- **Resource-Constrained Applicability:** Suitable for deployment on mobile devices and embedded systems.
- **Empirical Validation:** Comprehensive experiments across multiple language pairs confirm the quality-efficiency tradeoff.

## Limitations

- **Scalability:** Results may not generalize to all language pairs or very large multilingual settings.
- **No Pre-training:** Trained from scratch without transfer learning, unlike modern mBART or M2M-100 models.
- **Tokenization:** The paper does not explicitly describe the subword tokenization strategy (e.g., BPE, SentencePiece), affecting reproducibility.
- **Limited Long-Sequence Analysis:** No dedicated analysis of performance on very long sentences despite attention's theoretical advantage.
- **Single Attention Head:** Uses one attention head, whereas multi-head attention in Transformers captures multiple alignment types simultaneously.

# Model Architecture and Implementation

## BiLSTM Encoder

The encoder processes the source sentence in both forward and backward directions simultaneously. A key innovation from the paper is the use of learnable initial states: the hidden state  $h_0$  and cell state  $c_0$  are `nn.Parameter` objects updated during training, allowing the model to begin encoding from an optimised starting configuration rather than all-zeros.

The BiLSTM produces outputs of shape `[src_len, 1, 2×hidden_dim]`. The final forward and backward hidden states are merged via a linear projection layer to produce a single `[1, hidden_dim]` vector used to initialise the decoder.

```
self.h0 = nn.Parameter(torch.zeros(n_layers*2, 1, hidden_dim)) # learnable
self.rnn = nn.LSTM(embed_dim, hidden_dim, n_layers, bidirectional=True)
self.fc_h = nn.Linear(hidden_dim*2, hidden_dim) # merge fwd+bwd
```

## Bahdanau Attention Mechanism

The attention module computes a dynamic context vector at each decoder step using three learned linear transformations:  $W_1$  projects encoder states,  $W_2$  projects the decoder hidden state, and  $v$  collapses the tanh output to a scalar score per source token. Scores are normalized via softmax and the context vector is the weighted sum of encoder outputs.

```
energy = torch.tanh(self.W1(enc_2d) + self.W2(dec_2d)) # [src_len, hid]
scores = self.v(energy) # [src_len, 1]
attn_weights = torch.softmax(scores, dim=0)
context = (attn_weights * enc_2d).sum(dim=0, keepdim=True)
```

## Attention Decoder

At each decoding step the attention decoder: embeds the previous output token, calls the attention module to compute the current context vector, concatenates `[embedding, context]` as LSTM input, and produces logits from a three-way projection over `[LSTM output, context, embedding]`. This gives the model explicit access to all three signals at every prediction step.

## Baseline: No-Attention Decoder

The baseline decoder uses the same BiLSTM encoder but replaces dynamic attention with a single fixed context vector computed as the mean of all encoder outputs before decoding begins. This vector is reused unchanged at every decoder step, recreating the classic information bottleneck that attention is designed to eliminate.

```
context = enc_out.squeeze(1).mean(dim=0, keepdim=True) # computed once
```

## Training Setup

Both models are trained under identical conditions for a fair comparison:

| Hyperparameter      | Value                                    |
|---------------------|------------------------------------------|
| Embedding Dimension | 64                                       |
| Hidden Dimension    | 128 (encoder output = 256 due to BiLSTM) |

| Hyperparameter        | Value                                 |
|-----------------------|---------------------------------------|
| Number of Layers      | 1                                     |
| Dropout               | 0.3                                   |
| Learning Rate         | 0.001 (Adam optimizer)                |
| Epochs                | 30                                    |
| Gradient Clipping     | max norm = 1.0                        |
| Teacher Forcing Ratio | 0.5 (train) / 0.0 (eval)              |
| Loss Function         | CrossEntropyLoss (PAD index ignored)  |
| Dataset               | Tatoeba EN-FR, 500 pairs, 80/20 split |

## Comparison: With vs. Without Attention

### Model Modifications

#### With Attention — BiLSTM + Bahdanau

The attention model uses the full architecture as described in the paper. At each decoder step  $t$ , a context vector is computed dynamically from all encoder states. The decoder input is  $[\text{embed}(y_{t-1}), c_t]$  and logits are produced from  $[\text{LSTM\_out}, c_t, \text{embed}(y_{t-1})]$ . Total trainable parameters: 787,811.

#### Without Attention — Vanilla Seq2Seq Baseline

The no-attention model uses the same BiLSTM encoder but the decoder receives a fixed context vector (mean of encoder outputs) at every step regardless of what token is being generated. This is the classical information bottleneck. Total trainable parameters: 604,451.

### Quantitative Comparison

| Metric                          | With Attention               | Without Attention         |
|---------------------------------|------------------------------|---------------------------|
| Final Train Loss                | 0.5092                       | 0.6195                    |
| Final Validation Loss           | 5.4683                       | 4.5422                    |
| Val Perplexity (lower = better) | 237.05                       | 93.90                     |
| Avg Unigram BLEU Score          | 0.4896                       | 0.4479                    |
| Training Time (seconds)         | 286.2 s                      | 190.1 s                   |
| Trainable Parameters            | 787,811                      | 604,451                   |
| Context Type                    | Dynamic (per-step attention) | Fixed (mean vector)       |
| Dataset                         | Tatoeba EN-FR (500 pairs)    | Tatoeba EN-FR (500 pairs) |

### Translation Output Samples

Side-by-side translation comparison on 8 test sentences from the held-out set:

| Source (EN)            | Reference (FR)             | With Attention            | Without Attention         |
|------------------------|----------------------------|---------------------------|---------------------------|
| Did you miss me?       | Je t'ai manqué ?           | je <unk> manqué ?         | je <unk> manqué ?         |
| Did you miss me?       | T'ai-je manqué ?           | je <unk> manqué ?         | je <unk> manqué ?         |
| Did you miss me?       | Vous ai-je manqué ?        | je <unk> manqué ?         | je <unk> manqué ?         |
| Are they all the same? | Sont-ils tous identiques ? | sont-ils tous les <unk> ? | sont-ils tous les <unk> ? |
| Are they all the same? | Sont-ils tous les memes ?  | sont-ils tous les <unk> ? | sont-ils tous les <unk> ? |
| Thank you very much!   | Merci beaucoup !           | merci beaucoup !          | merci <unk> !             |

| Source (EN)                 | Reference (FR)                       | With Attention                       | Without Attention                    |
|-----------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
| Thank you very much!        | Merci vraiment.                      | merci beaucoup !                     | merci <unk> !                        |
| Where are the eggs, please? | S'il vous plait, ou sont les oeufs ? | s'il vous plait, ou sont les <unk> ? | s'il vous plait, ou sont les <unk> ? |

The attention model correctly generates 'beaucoup' where the no-attention model produces <unk>, demonstrating the benefit of dynamic source alignment even on a small vocabulary.

## Visualizations

### Training and Validation Loss Curves

Both models show decreasing training loss across 30 epochs. The attention model reaches a lower final training loss (0.51 vs 0.62), indicating a stronger learning signal from the dynamic context. Validation loss is higher for the attention model due to overfitting on the 500-pair dataset, which is discussed in the analysis section.

### Attention Heatmaps

The attention heatmaps for test sentences show near-diagonal alignment patterns. For 'the cat is sleeping', decoder steps generating 'le' attend to 'the', steps generating 'chat' attend to 'cat', and steps generating the verb attend to 'sleeping'. This structured alignment confirms the model has learned correct EN-FR word correspondences without explicit supervision.

## Result Analysis and Discussion

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### How Attention Improves Performance

**Training Loss:** The attention model achieves a lower final training loss (0.51 vs 0.62), indicating that dynamic context vectors provide a richer learning signal. By attending to relevant source tokens at each step, the model receives more precise gradient updates throughout training.

**BLEU Score:** The attention model outperforms the baseline by approximately 9.3% relative improvement (0.4896 vs 0.4479), demonstrating better word-level overlap with reference translations. This improvement is consistent with the paper's findings at WMT14 scale.

**Alignment and Context Understanding:** Attention allows the decoder to focus on the most relevant source tokens at each generation step. When generating a French subject, the attention weights highlight the English subject tokens; when generating the verb they shift to the verb region. This dynamic alignment is impossible in the no-attention model which sees only a single averaged vector throughout decoding.

**Performance on Longer Sequences:** Although this experiment used short sentences (max 15 tokens), attention's benefit becomes more pronounced as sequence length increases. The fixed-context model suffers increasing information loss as source length grows, while attention always has direct access to all encoder states.

### Limitations Without Attention

- **Fixed Context Vector Bottleneck:** All source information is compressed into a single mean vector, causing progressive information loss with longer inputs.
- **Higher Training Loss:** Converges to 0.62 vs 0.51, indicating weaker source-target alignment learning.
- **Lower BLEU Score:** 9.3% relative drop confirms lower output quality against reference translations.
- **No Token-Level Alignment:** Cannot learn which source tokens correspond to which target tokens.

### Anomalous Validation Results

The no-attention model achieves lower validation loss (4.54 vs 5.47) and lower perplexity (93.90 vs 237.05) despite having a lower BLEU score. This apparent contradiction arises because the attention model has 30% more parameters. On only 400 training pairs, these additional parameters overfit more aggressively to the training distribution, leading to higher cross-entropy on the validation set.

However, BLEU score tells a different story: despite the lower validation cross-entropy, the no-attention model produces translations with lower word-level overlap with references. This is a well-known discrepancy in NLP: cross-entropy loss does not always correlate with human-perceived translation quality. BLEU is the more appropriate metric here. At WMT14 scale both metrics improve together, as the paper demonstrates.

### Why Attention Gives Better Results

- **Better gradient flow:** Direct differentiable paths from decoder step  $t$  to encoder position  $i$  allow gradients to reach relevant source tokens without traversing the full LSTM chain.



- **Word alignment learning:** Attention weights learn meaningful linguistic correspondences visible in heatmaps (e.g., English 'cat' aligns to French 'chat').
- **Scalability to long sequences:** Attention degrades gracefully with length; fixed-context degrades rapidly since its vector is fixed-size regardless of input length.

## Cases Where Attention Shows Significant Improvement

- **Long sentences:** Sentences with 10+ tokens benefit most as the fixed context vector becomes too compressed to represent them accurately.
- **Syntactically different language pairs:** EN-FR word order differences (adjective placement) require the decoder to attend to non-adjacent source positions.
- **Idiomatic expressions:** Multi-word English phrases mapping to a single French word require multi-source alignment, which only attention can do.

## Comparison with Paper Results

Wu and Xing (2024) report competitive BLEU scores on WMT14 EN-FR at scale, outperforming Transformer baselines under equivalent storage constraints. The small-scale Tatoeba experiment cannot replicate the absolute numbers but correctly reproduces the qualitative trend: attention produces better translations (higher BLEU) than the no-attention baseline, confirming the paper's core architectural claim.

# Conclusion

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## Key Findings

This project studied, implemented, and experimentally compared the BiLSTM-Attention Seq2Seq architecture from Wu and Xing (2024) against a vanilla Seq2Seq baseline. The key findings are:

- The attention model achieves lower final training loss (0.51 vs 0.62) and higher BLEU score (0.49 vs 0.45) on the Tatoeba EN-FR dataset.
- Bahdanau additive attention enables dynamic source alignment at each decoding step, resolving the fixed-context bottleneck.
- The attention model requires approximately 50% more training time (286s vs 190s) and 30% more parameters — the cost of the quality improvement.
- On a small 500-pair dataset, the attention model overfits more aggressively (higher val loss), but BLEU remains the more appropriate translation quality metric where attention retains its advantage.
- Learnable initial hidden/cell states allow the encoder to begin from an optimised starting configuration rather than all-zeros, a practical innovation from the paper.

## Importance of the Attention Mechanism

The attention mechanism represents one of the most significant advances in sequence-to-sequence learning. Prior to attention (Bahdanau et al., 2015), all Seq2Seq models suffered from the fixed-context bottleneck: the entire source sentence had to be compressed into a fixed-size vector. Attention solved this by allowing the decoder to selectively read any part of the encoder output at each step — a soft, differentiable search over source positions.

This mechanism directly inspired the Transformer architecture (Vaswani et al., 2017), which extends the idea to self-attention across all sequence positions. In this sense, attention is not merely an improvement to RNN models — it is the conceptual foundation of modern large language models. Wu and Xing (2024) demonstrate that attention-augmented RNN models remain relevant and competitive even in the Transformer era, particularly under storage and deployment constraints.

## Real-World Applicability

- **Mobile and edge translation:** Smaller model size makes it deployable on smartphones and IoT devices with limited memory.
- **Low-resource language translation:** Effective with limited training data, suitable for language pairs lacking millions of parallel sentences.
- **Streaming / real-time translation:** Sequential step-by-step decoding is compatible with streaming inference where output tokens are generated as source tokens arrive.
- **Educational and research use:** The interpretable architecture is an excellent baseline for studying translation behavior and attention alignment patterns.

In summary, this project successfully reproduced the core architectural principles of Wu and Xing (2024), verified the advantage of attention over the no-attention baseline through quantitative metrics and attention heatmap visualizations, and demonstrated that the BiLSTM-Attention model represents a principled, efficient, and practically deployable approach to neural machine translation.

## References

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